JEE APEX FTS – JEE ADV. PART TEST-01 [P1]

KEY SHEET

MATHEMATICS

1	**CD**	2	**BC**	3	**ABCD**	4	**A**	5	**A**	6	**D**
7	**C**	8	**1**	9	**2**	10	**1**	11	**18**	12	**1**
13	**4**	14	**B**	15	**C**	16	**C**	17	**C**		

PHYSICS

18	**BC**	19	**AB**	20	**ABD**	21	**D**	22	**C**	23	**B**
24	**A**	25	**4**	26	**6**	27	**3**	28	**6**	29	**8**
30	**8**	31	**A**	32	**C**	33	**C**	34	**C**		

CHEMISTRY

35	**ABCD**	36	**CD**	37	**ABC**	38	**C**	39	**A**	40	**C**
41	**A**	42	**4**	43	**4**	44	**5**	45	**6**	46	**36**
47	**10**	48	**D**	49	**C**	50	**B**	51	**B**		

SOLUTIONS

MATHEMATICS

01. $f(x) = 1 - x$

02. $A.adj(2B) = 16I$

$A(4adjB) = 16I \Rightarrow A.adjB = 4I \Rightarrow A|B|.B^{-1} = 4I$

$AB^{-1} = 2I \Rightarrow A = 2B$

$B.adjA = B.adj(2B) = 4B.adjB = 4 \mid B \mid I_3 = 8I_3$

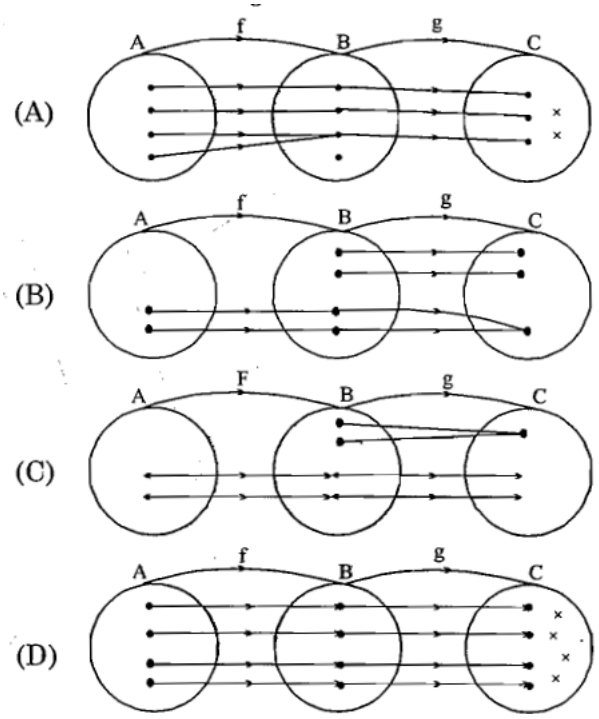
$A.adjB = 4I \Rightarrow A^{-1}A.adjB \Rightarrow adjB = 4I \Rightarrow A^{-1}.AadjB = 4A^{-1} \Rightarrow adjB = 4A^{-1}$

$A^{-1}adjB = A^{-1}.(4A^{-1}) = 4(A^{-1})^2$

$(A^{-1}(adjB))^{-1} = (4A^{-1}A^{-1})^{-1} = \frac{1}{4}A^2 = \frac{1}{4}(2B)^2 = B^2$

03. (A) We have $f : A \to B$

$g : B \to C$ and $gof : A \to C$



04. $|\cos 2x| = |\sin x| \Rightarrow 2\cos^2 2x = 2\sin^2 x = 1 - \cos 2x$

$\Rightarrow (\cos 2x - 1)(\cos 2x + 1) = 0 \Rightarrow \cos 2x = -1 \text{ or } \frac{1}{2}$

$\Rightarrow x = \frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$. So 3 solutions

05. Let $S = \sum_{k=1}^{\infty} \frac{P_k}{3^k} = \frac{P_1}{3} + \frac{P_2}{3^2} + \frac{P_3}{3^3} + \frac{P_4}{3^4} + \frac{P_5}{3^5}$

$S = \frac{1}{3} + \frac{1}{3^2} + \frac{2}{3^3} + \frac{3}{3^4} + \frac{5}{3^5} + \frac{8}{3^6} + ... + \frac{P_n}{3^n}$

$S = \frac{1}{3^2} + \frac{1}{3^3} + \frac{2}{3^4} + \frac{3}{3^5} + ... + \frac{P_{n-1}}{3^n} + ...$

$\frac{S}{9} = \frac{1}{3^3} + \frac{1}{3^4} + \frac{2}{3^5} + ... + \frac{P_{n-2}}{3^n}$

$S - \left(\frac{S}{3} + \frac{S}{9}\right) = \frac{1}{3} \Rightarrow S = \frac{3}{5}$

06. $f(\theta)=\dfrac{(\sin 7\theta+\sin 5\theta)+5(\sin 5\theta+\sin 3\theta)+12(\sin 3\theta+\sin\theta)}{\sin 6\theta+5\sin 4\theta+12\sin 2\theta}$

$=\dfrac{2\cos\theta(\sin 6\theta+5\sin 4\theta+12\sin 2\theta)}{\sin 6\theta+5\sin 4\theta+12\sin 2\theta}=2\cos\theta$

07. $\dfrac{x}{1}=\dfrac{y+2}{1}=\dfrac{z}{1}=\lambda$ .....(1)

$\dfrac{x+2}{2}=\dfrac{y}{1}=\dfrac{z}{1}=\mu$ .....(2)

Any point of line (1) is $(\lambda,\lambda-2,\lambda)$

Any point of line (2) IS $(2\mu-2,\mu,\mu)$

$\dfrac{\lambda-2\mu+2}{2}=\dfrac{\lambda-2-\mu}{1}=\dfrac{\lambda-\mu}{2}$

$\lambda=6,\ \mu=2$

08. Let $f(x)=x^4.3^{|x-2|}.2^{|x-5|}.5^{x-1}$

Now, $|f(x)|=-f(x)\ (Given)$

$\Rightarrow f(x)\le 0$

$\Rightarrow x^4.3^{|x-2|}.2^{|x-5|}.5^{x-1}\le 0$

Which is possible when $x^4=0$

$\therefore\ x=0$

09. As $\lim\limits_{x\to\pm\infty} f(x)=\infty$

$\therefore$ degree of $g(x)$ must be less than 4.

As $\lim\limits_{x\to-1} f(x)=6\Rightarrow \lim\limits_{x\to-1}\dfrac{(x^2-1)(x^2-4)}{g(x)}=6\ \Rightarrow g(x)$

Must have $(x+1)$ as a factor.

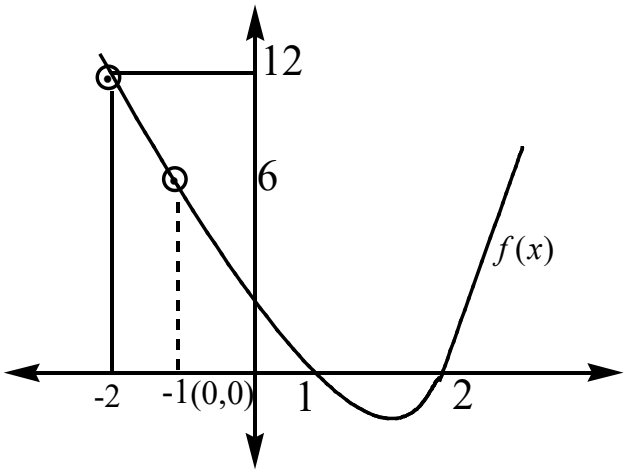
Again, $\lim\limits_{x\to-2} f(x)\ne 0$ but finite $\Rightarrow g(x)$ must

Have $(x+2)$ another factor.

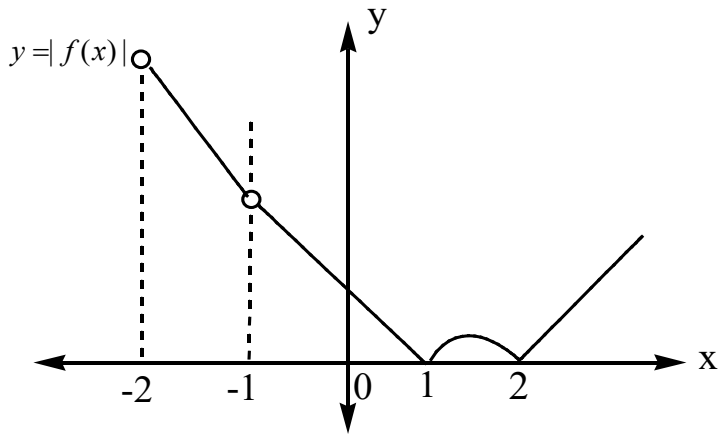
$\therefore g(x)=\lambda(x+1)(x+2)$

Now, $\lim\limits_{x\to-1}\dfrac{(x^2-1)(x^2-4)}{\lambda(x+1)(x+2)}=6\ \Rightarrow \lim\limits_{x\to-1}\dfrac{(x-1)(x-2)}{\lambda}=6\Rightarrow\dfrac{(-2)(-3)}{\lambda}=6\Rightarrow\lambda=1$

Graph of $f(x)$**:** $f(x)=\dfrac{(x^2-1)(x^2-4)}{(x+1)(x+2)}\ x\in R-\{-2,-1\}\ =(x-1)(x-2)\ x\in R-\{-2,-1\}$



(C) $y=|f(x)|$



Clearly $y=|f(x)|$ is continuous $\forall\ x\in$

Domain hence $L=0$ and non derivable at $x=1$ and $x=2$

$\therefore L+M=0+2=2$

10. $f(x)=\sin^{-1}(x+2019)-\cos^{-1}(x+2021)$

$-1\le x+2019\le 1$ $\qquad$ $-1\le x+2021\le 1$

$-2020\le x\le -2018$ $\qquad$ $-2022\le x\le -2020$

$\therefore$ Domain of $f(x)$ is $\{-2020\}$

$g(x)=\sin^{-1}(x+2017)+\cos^{-1}(x+2019)$

$-1\le x+2017\le 1$ $\qquad$ $-1\le x+2019\le 1$

$-2018\le x\le -2016$ $\qquad$ $-2020\le x\le -2018$

$\therefore$ Domain of $f(x)$ is $\{-2018\}$

$x_1=-2020$ and $x_2=-2018$

$$\left|\lim_{x\to -4038}\frac{\sin(x+4038)-\sin(2x+8076)}{\tan(x+4038)}\right|=1$$

11. We have

$\vec{V}_1.\vec{V}_2=a+b+c=0$

But $a,b,c\in\{-2,-1,0,1,2\}$

Now,

(i) if $a=1,b=-1,c=0$, number $=3!=6$

(ii) if $a=2,b=-2,c=0$, number $=3!=6$

(iii) if $a=1,b=1,c=-2$, number $=\dfrac{3!}{2!}=3$

(iv) if $a=-1,b=-1,c=2$, number $=\dfrac{3!}{2!}=3$

$\therefore$ Total $=18$

12. $f(x)=x^2+bx+c\forall x\in R$

Least value at $\dfrac{-b}{2}=-1\Rightarrow b=2$

Graph of $f(x)$ cuts y-axis, when $x=0$

$\Rightarrow c=2$

$\Rightarrow f(x)=x^2+2x+2$

Least value of $f(x)=1$

13. $(a-1)x+0y+z=\alpha \quad ....(1)$

$x+(b-1)y+0z=\beta \quad ....(2)$

$0x+y+(c-1)z=\lambda \quad .....(3)$

For no unique solution D = 0

$$\begin{vmatrix}(a-1) & 0 & 1\\ 1 & (b-1) & 0\\ 0 & 1 & (c-1)\end{vmatrix}=0$$

$(a-1)(b-1)(c-1)+1=0$

$\therefore a=2;b=2;c=0$

Hence, $|a+b+c|=4$

14. $$\left.\begin{aligned}&\log_{10}(2xy)=4+(\log_{10}x-1)(\log_{10}y-2)\\&\log_{10}(2yz)=4+(\log_{10}y-2)(\log_{10}z-1)\\&\log_{10}(zx)=2+(\log_{10}z-1)(\log_{10}x-1)\end{aligned}\right\}S$$

Let

$\log x-1=A$

$\log y-2=B$

$\log_{10}z-1=C$

In 'S'

$$\begin{cases}4+AB=\log 2+A+1+B+2\Rightarrow(A-1)(B-1)=\log 2\\4+BC=\log 2+B+2+C+1\Rightarrow(B-1)(C-1)=\log 2\\2+AC=\quad\quad C+1+A+1\Rightarrow(A-1)(C-1)=1\end{cases}$$

$\Rightarrow(A-1)(B-1)(C-1)=\pm\log 2$

$$\left.\begin{array}{l} C-1=\pm 1 \\ A-1=\pm 1 \\ B-1=\pm\log 2 \end{array}\right\}$$

$(A,B,C)=(2,1+\log 2,2),(0,1-\log 2,0)$

$(x,y,z)=(10^3,2000,10^3),(10,500,10)$

15. $2x^3+x^2(3-7a)+x(3a^2-5a)+(3a^2+2a-1)=0$

$(x+1)(2x-(a+1))(x-(3a-1))=0$

$x=-1,\dfrac{a+1}{2},3a-1$

For exactly equal roots $-1=\dfrac{a+1}{2}\rightarrow a=-3$

$-1=3a-1\rightarrow a=0$ and $\dfrac{a+1}{2}=3a-1\Rightarrow a=\dfrac{2}{5}$

For all negative roots a + 1 < 0 and $a<\dfrac{1}{3}$ $\quad\therefore a<-1$

–1 lies between the roots

$\dfrac{a+1}{2}<-1, 3a-1>-1\Rightarrow a<-3$ and a > 0

$\therefore$ No value.

Or $\dfrac{a+1}{2}>-1, 3a-1<-1\Rightarrow -3<a<0$

All the roots lie in [–1, 2] if $-1\le\dfrac{a+1}{2}\le 2\ \Rightarrow -2\le a+1\le 4$

$-3\le a\le 3$ and $-1\le 3a-1\le 2\Rightarrow 0\le a\le 1$

$a\in[0,1]$

16. P) $-1\le[2x]\le 1$ and $[x]\ne 0$

$-1\le 2x\le 2$ and $x\notin[0,1)$

$-\dfrac{1}{2}\le x\le 1$ and $x\notin[0,1)$

$\therefore x\in\left[-\dfrac{1}{2},0\right)$

Q) $-2\le 2x+4\le 10\Rightarrow -6\le 2x\le 6\ \Rightarrow -3\le x\le 3$

R) $f(x)=x^4-x^2=x^2(x^2-1)$

$f(t)=t(t-1), 0\le t<\infty$

$\therefore f(t)\in\left[\dfrac{-1}{4},\infty\right)$

S) $f(x)=\log_2\left(1-\left[\sin^2 x\right]\right)+\cos^2 x$

$\left[\sin^2 x\right]<1\Rightarrow\left[\sin^2 x\right]=0\le\sin^2 x<1$

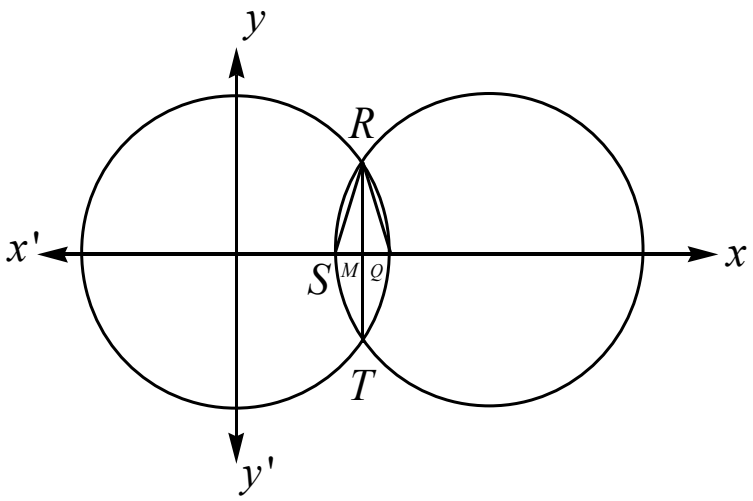
$\therefore \sin^2 x \neq 1 \Rightarrow 0 < \cos^2 x \leq 1 \;\; \therefore$ range of $f(x)$ is $(0,1]$

17. P) $y - y^2 = a^2$

$D > 0 \Rightarrow a \in \left(-\frac{1}{2}, \frac{1}{2}\right)$

$|\alpha - \beta| = 1$

Q)



Variable circle $(x-1)^2 + (y-0)^2 = r^2$.

$x^2 + y^2 - 2x + (1-r)^2 = 0; 0 < r < 2$.

$R: \left\{\frac{2-r^2}{2}, \frac{r\sqrt{4-r^2}}{2}\right\}$

$\Delta QRS = \frac{1}{2}\frac{r^2\sqrt{4-r^2}}{2}$

Maximum when $r = \sqrt{\frac{8}{3}}$

R) $AB = 2r\sqrt{\frac{S_1}{S_1 + r^2}} = 2.3\sqrt{\frac{1}{1+9}}$

S) $L_T = \sqrt{d^2 - (r_1 + r_2)^2} = 24$

PHYSICS

18. $\frac{dH}{dT} = b(T - T_0) = 3 \times 10^{-3} \times 2\pi \times 0.5 \times \ell \times 80 = 2.4\pi \times 10^{-11}$ Cal/sec

Case-2

$\frac{dH}{dt} = 3 \times 10^{-3} \times 1 \times \ell(T - 20) \times 2\pi = \frac{2\pi k\ell}{\ell n(2)} \times (100 - T)$

$$3(T-20)=\frac{2.8\times10^{-3}}{0.7}(100-T)$$

$$3T-60=400-4T$$

$$7T=460$$

$$T=\frac{460}{7}=65.7^{\circ}C$$

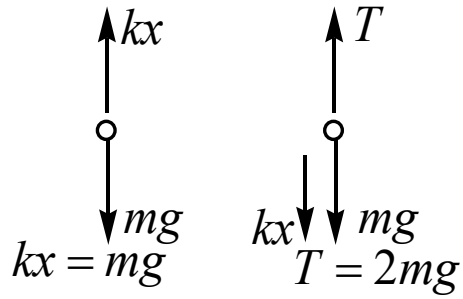
$$\frac{\frac{dH'}{dt}}{\frac{dH}{dt}}=\frac{3\times10^{-3}\times\ell\left(\frac{460}{7}-20\right)\times2\pi}{2.4\pi\times10^{-1}\ell}$$

$$=\frac{1}{40}\left(\frac{460-140}{7}\right)=\frac{320}{280}$$

19. Relative velocity between the adjacent particles is v along the side of square. If area becomes half, length becomes $\frac{l}{\sqrt{2}}$. Therefore the separation is decreased by

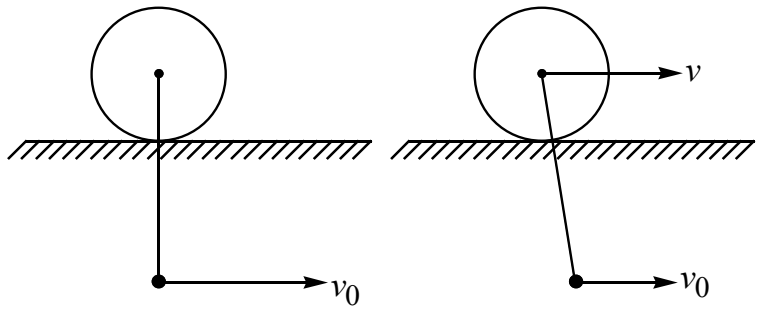
$$\left(1-\frac{1}{\sqrt{2}}\right)l. \quad \therefore t=\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)\frac{1}{v}$$

20. Initially in equilibrium



When thread in cut, no change in spring force immediately i.e. $a_B=0$ and $a_A=2g\downarrow$

21.



$$mV_0=(M+m)V$$

$$V=\frac{mV_0}{M+m}$$

From energy conservation

$$\frac{1}{2}mV_0^2+Mg\ell=mgh+Mg\ell+\frac{1}{2}(M+m)V^2$$

$$\frac{1}{2}mV_0^2 - \frac{1}{2}\frac{m^2V_0^2}{(M+m)} = mgh$$

$$\frac{1}{2}\frac{mV_0^2[M+m-m]}{[M+m]} = mgh$$

$$h = \frac{1}{2}\frac{V_0^2}{g}\frac{M}{(M+m)}$$

22. $\because$ Given $\frac{dv}{dt} = \frac{v^2}{R}$.....(1)

$$\Rightarrow \int_{u_0}^{v}\frac{dv}{v^2} = \int_0^t \frac{1}{R}dt \Rightarrow R\left(\frac{1}{u_0} - \frac{1}{v}\right) = t.....(2)$$

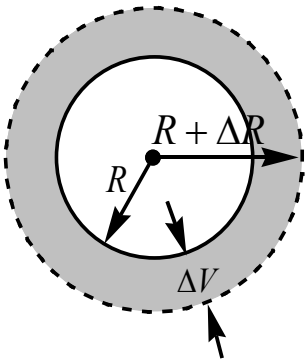
Again from equation (1) $\because \frac{dv}{ds}.\frac{ds}{dt} = \frac{v^2}{R} \Rightarrow v\frac{dv}{ds} = \frac{v^2}{R}$

$$\int_{u_0}^{v}\frac{dv}{v} = \int_0^{2\pi R}\frac{ds}{R} \Rightarrow v = u_0e^{-2\pi}.....(3)$$

From (2) and (3) $t = \frac{R}{u_0}\left(1 - e^{-2\pi}\right)$

23. $\theta = \tan^{-1}\left(\frac{a}{g}\right)$

24.



According to the problem, mass of water (m) = 100g

Change in temperature $\Delta T = 0 - (-10) = 10^\circ C$

Specific heat of water $(S_w) = 1cal\ /\ g\ /^\circ C$

Latent heat of fusion of water $L^w_{fusion} = 80cal\ /\ g$

Amount of heat required to change the temperature of 100g of water at $-10^\circ C$ to $0^\circ C$

$Q = mS_w\Delta T \quad = 100 \times 1 \times [0 - (-10)] = 1000\,cal$

Let m gram of ice be melted. $\therefore Q = ml$ Or $m = \frac{Q}{L} = \frac{1000}{80} = 12.5g$

25. Let length of semi major axis is a and eccentricity is e.

$$-\frac{GMm}{R_e}+\frac{1}{2}m\left(\sqrt{\frac{GM_e}{R_e}}\right)^2=\frac{GM_em}{2a}\Rightarrow a=R_\circ$$

As the separation between ends of semi minor axis and focus is equal to length of semi major axis we can say that launching sites ends of semi minor axis.

$\Rightarrow R_e\cos 45^\circ = ae \Rightarrow e = 1/\sqrt{2}$

Then maximum separation will be $a+ae=R_e\dfrac{\left(\sqrt{2}+1\right)}{\sqrt{2}}$

26. For any small change of pressure dp, there will be a change of volume dV and $dp=-B\dfrac{dV}{V}$. In this change, work is done on the system and the energy stored in the materials is $dW=-pdV=\left(\dfrac{V}{B}\right)pdp$

In the change mentioned in the question, the total work done is

$$W=-\int_{V_i}^{V_f}pdV=\int_{V_i}^{V_f}\frac{V}{B}pdp$$

The change in volume is negligible and volume can be treated as constant

$$W=\frac{V_0}{B}\int_{p_0}^{p}pdp=\frac{V_0}{B}\left(\frac{p^2-p_0^2}{2}\right)$$

Extra energy stored per unit volume is $\dfrac{W}{V_0}=\dfrac{1}{2B}\left(p^2-p_0^2\right)$

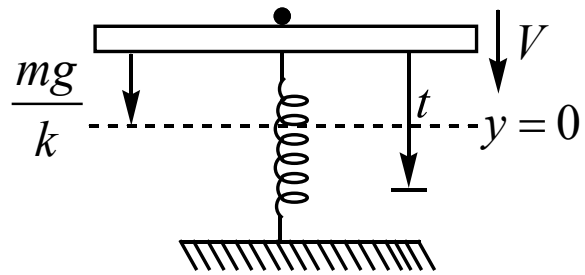
27. Velocity of the particle just before collision

$$u=\sqrt{2g\times\frac{4.5mg}{K}}\quad\Rightarrow\quad u=3g\sqrt{\frac{m}{K}}$$

Now it collides with the plate

Now just after collision velocity (V) of system plate of "plate + particle"

$$mu+3mV\quad\Rightarrow\quad V=\frac{u}{3}=g\sqrt{\frac{m}{K}}$$



Now system perform's SHM with time period $T=2\pi\sqrt{\dfrac{3m}{K}}$ and mean position as $\dfrac{mg}{K}$.

distance below the point of collision.

Let the equation of motion be $y=A\sin(\omega t+\phi)$

for t = 0 y = mg/K

$\frac{mg}{K} = A\sin\phi$ .....(1)

Now for amplitude $V = \omega\sqrt{A^2 - x^2}$

$g\sqrt{\frac{m}{K}} = \sqrt{\frac{K}{3m}}\sqrt{A^2 - \frac{m^2g^2}{K^2}} \quad \Rightarrow \quad \left(\sqrt{3}\frac{mg}{K}\right)^2 = A^2 = \frac{m^2g^2}{K^2} \Rightarrow A = \frac{2mg}{K}$ .....(2)

By (1) & (2)

$x = \frac{A}{2}$ to x = 0 $\Rightarrow t = \frac{T}{12}$

X = 0 to x = A $\Rightarrow t = T/4$

Total time $= \frac{T}{12} + \frac{T}{4} = \frac{2\pi}{3}\sqrt{\frac{3m}{K}}$

28. $\left(P_0 + \frac{4S}{R}\right)\left(\frac{4}{3}\pi R^3\right) = \left(\frac{P_0}{2} + \frac{4S\times 4}{5R}\right)\times\frac{4}{3}\pi\left(\frac{5R}{4}\right)^3$

$\Rightarrow P_0 + \frac{4S}{R} = \frac{125P_0}{128} + \frac{25S}{4R} \quad \Rightarrow \quad P_0 = \frac{96S}{R} = \frac{16nS}{R}$

$\therefore n = 6$

29. $a_x = \frac{6}{1+5} = 1\ ms^{-2}$

$N = 1\times 1 = 1N$

$f_r = \mu N = \frac{1}{2}$

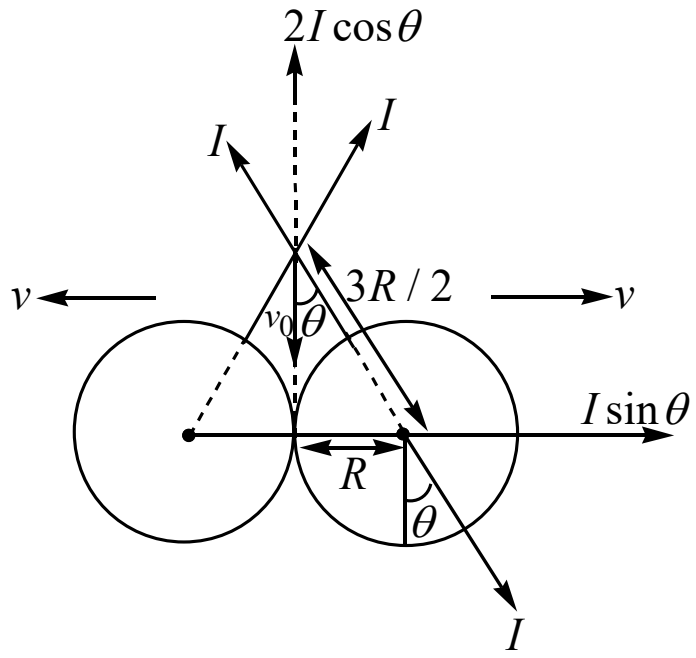
$a = \frac{fr}{m} = \frac{1}{2}$

$0 = v_0 - \frac{1}{2}\times t$

T = 10 sec.

30. I = Impulse by each of bigger balls on smaller ball $\sin\theta = 2/3, \cos\theta = \frac{\sqrt{5}}{3}$ By impulse momentum theorem, for smaller ball $2I\cos\theta = Mv_0 \Rightarrow I = \frac{3Mv_0}{2\sqrt{5}}$ Now, $I\sin\theta = Mv$ (here v = speed acquired by each of the bigger balls after collision) applying law of conservation of energy, we get

$\frac{1}{2}M\left(\frac{v_0}{\sqrt{5}}\right)^2 = \frac{1}{2}kx^2 \Rightarrow x = v_0\sqrt{\frac{M}{5K}} \Rightarrow x = 1.6\sqrt{\frac{1.25}{5\times 100}} \Rightarrow x = \frac{1.6}{20}m \Rightarrow x = 8\ cm$

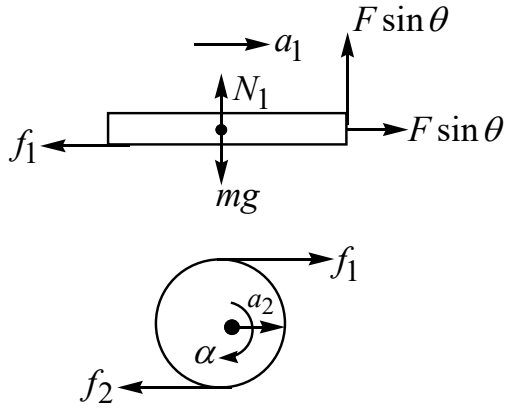


31. The FBD & various parameters are shown in figure. From various dynamics equation.

$$F\cos\theta - f_1 = 3ma_1$$

$$f_1 - f_2 = Ma_2$$

$$\alpha = \frac{(f_1 + f_2)R}{\frac{1}{2}MR^2}$$



Solving above equation we get

$$a_2 = \frac{4F\cos\theta}{[3M+8m]}, f_1 = \frac{3MF\cos\theta}{[3M+8m]}$$

$$a_1 = \frac{8F\cos\theta}{[3M+8m]}, f_2 = \frac{MF\cos\theta}{[3M+8m]}$$

32. Total change in KE of block is zero. Work done by friction on block w.r.t. belt
Distance slid by the block on the belt relative to belt
Change in momentum of block relative to belt

33. $\Rightarrow$For option (P) temperature is minimum hence λm will be maximum $\beta = \frac{\lambda D}{d} \Rightarrow \beta$ will also be maximum $\Rightarrow$For option (Q) T = 3000

$$\lambda m = \frac{b}{T} = \frac{2.9\times10^{-3}}{30000}$$

$$\lambda m = \frac{2.9}{3}\times10^{-6} = 0.96\times10^{-6} = 966.6\ nm$$

$$P_{3000} = 6A(3000)^4$$

$$P_{6000} = 6A(6000)^4$$

$$\frac{P_{3000}}{P_{6000}} = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

$$P_{3000} = \frac{1}{16}P_{6000}$$

$Q-4$ $\Rightarrow$For (R) T = 5000 K

$$\lambda m = \frac{2.9\times10^{-3}}{5\times10^{3}} = 0.58\times10^{-6} = 580 \text{ nm}$$

Visible to human eyes

R – 2

$\Rightarrow$for (S) T = 10,000 $\rightarrow$maximum

Hence (3) is wrong as it has minimum (λm)

34. (I) $V_{BA}^2 = V_A^2 + V_B^2 - 2V_{AB}\cos\theta$

AS $\omega_A = \omega_B, \theta = 90^\circ$ remains constant

Also, $V_A = V_B = 1m/s$ So, $V_{BA} = \sqrt{2}\ m/s$

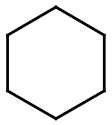
(II) $\vec{u}_A - = \frac{5\pi}{2}\hat{i} + \frac{5\pi}{2}\hat{j}$

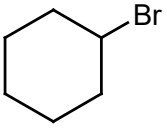
$$\vec{V}_A - = \frac{5\pi}{2}\hat{i} + \left(\frac{5\pi}{2} - 10.\frac{\pi}{3}\right)\hat{j} = \frac{5\pi}{2}\hat{i} - \frac{5\pi}{6}\hat{j}$$

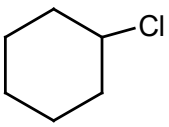
$$\vec{u}_B - = -\frac{5\pi}{2}\hat{i} + \frac{5\pi}{2}\hat{j}$$

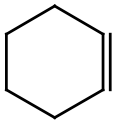
CHEMISTRY

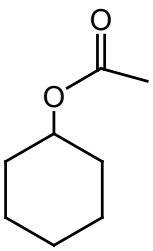
35.

P = 

S = 

Q = 

R = U = 

T = 

36. CONCEPTUAL

37. (i) In H_2^+, the removal of one electron from H_2 results in a lower BDE and increased BL.

H_2^+: BDE = 255 kJ/mol and bond length = 106 pm

H_2: BDE = 436 kJ/mol and bond length = 74 pm

(ii) Groups 7,8,9 – hydride gap

In group 6, only Cr form metallic hydride

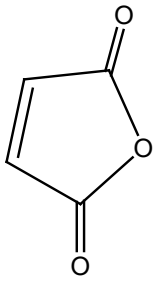
(iii) H: covalent radius = 37 pm ; H^-: ionic radius = 208 pm

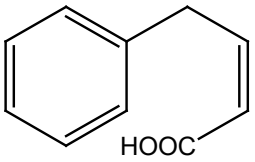
(iv) F : covalent radius = 64 pm; F^-: ionic radius = 133 pm

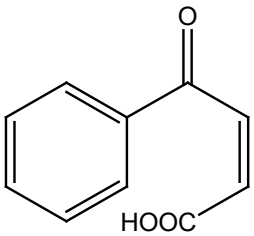
(v) Li :Metallic radius = 152 pm; Li^+ : ionic radius = 76 pm

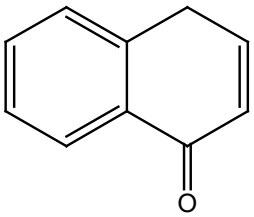
(vi) Mg: Metallic radius = 160 pm; Mg^{2+} : ionic radius = 72 pm

38.

P = 

R = 

Q = 

T = 

39. $C_2H_2 + 2.5O_2 \rightarrow 2CO_2 + H_2O$

2 *mole* 5 *mole*

$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$

1 *mole* 3 *mole*

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

2 *mole* 4 *mole*

Total weight of the hydrocarbon mixture

$= 2 \times 26 + 1 \times 28 + 2 \times 16$

$= 112\ gram$

Total mole of $O_2 = 5 + 3 + 4 \Rightarrow 12$

Mole of air required $= 12 \times 5$

$= 60$

Wt. of air required $= 60 \times 28.8$

$= 1728\ gram$

Wt. of air required per gram of the hydrocarbon mixture $= \frac{1728}{112} gram$

40. For a given group 1 metal, LE is : $O^{2-} < O_2^- < O_2^{2-}$

41. (i) $B_2O_3 + 3CaF_2 + 3H_2SO_4 \rightarrow 2BF_3 + 3CaSO_4 + 3H_2O$

(ii) BF_3 fumes in moist air and is partially hydrolysed by water

(iii) TlI_3 is actually thallium(I) triiodide

(iv) Extent of $\pi-$bonding: $NO_3^- > CO_3^{2-} > BO_3^{3-}$

42. (i), (iv),(vii),(viii)

43. A,E,F,G

44. For $Cl_2(g)$

$$P = \frac{nRT}{V}$$

$$P = \frac{10 \times 0.082 \times 300}{164}$$

$$P = 1.5\ atm$$

$$Br_2(l) + Cl_2(g) \rightleftharpoons 2BrCl(g)$$

0.25 atm 1.5 atm 0

0.25 atm $1.5(1-\alpha)$ $2 \times 1.5 \times \alpha -$ at eq.

$$P\ Total = 0.25 + 1.5(1-\alpha) + 3\alpha$$

$$2.25 = 0.25 + 1.5 + 1.5\alpha$$

$$\frac{0.5}{1.5} = \alpha \quad \Rightarrow \alpha = \frac{1}{3}$$

Mole of Cl_2 converted to product $= \frac{10}{3}$

So mole of $Br_2(l)$ converted to product $\frac{10}{3}$.

To maintain 0.25 atm pressure of Br_2 vapours the no of moles required

$$n = \frac{PV}{RT} = \frac{0.25 \times 164}{0.082 \times 300} = \frac{5}{3}$$

So minimum mole of $Br_2(l)$ required $\Rightarrow$ Initially $\frac{10}{3} + \frac{5}{3} = 5$

45. $$Z = \frac{1}{4}.u_{aV}.N^* = \frac{1}{4}\sqrt{\frac{8RT}{\pi M}}.\frac{P}{kT} = \text{Constant}$$

or, $\frac{P}{\sqrt{T}} = \text{Constant} \Rightarrow P.V^{-1} = \text{ Constant}$

$$\therefore C_m = C_{V_1 m} + \frac{R}{1-x} = \frac{5}{2}R + \frac{R}{1-(-1)} = 3R = 6\ cal/K\ mol$$

46. $Kr: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

d-electron = 10

p-electron = 18

s-electron = 8

47. $2Fe^{+2} + 2H^{+} + H_2O_2 \rightarrow 2Fe^{+3} + 2H_2O \ (x = 2)$

$2MnO_4^{-} + 6H^{+} + 5H_2O_2 \rightarrow 2Mn^{+2} + 8H_2O + 5O_2 \ (y = 5)$

$2MnO_4^{-} + 3H_2O_2 \rightarrow 2MnO_2 + 3O_2 + 2H_2O + 2OH^{-} \ (z = 3)$

48. $NaNH_2$ will not form salt with alkene

Na metal cannot make salt with alkene

49. CONCEPTUAL

50. CONCEPTUAL

51. $Be_2H_4 : Be : Sp^2$

$B_2H_6 : B : Sp^3$

$Al_2(CH_3)_6 : Al : Sp^3$, bridge carbon: Sp^3

$Be_2(CH_3)_4 : Be : Sp^2$, bridge carbon: Sp^3